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This page tests your understanding of core machine learning (ML) concepts through interactive questions. It covers fundamental ML topics such as predictive power of features, supervised and unsupervised learning, and model training and evaluation. You'll learn how to choose the right ML approach for different problems and assess the effectiveness of a trained model. Links to further resources are provided to deepen your understanding of ML and its practical applications. The following questions help you solidify your understanding of core ML concepts.

Predictive power

Supervised ML models are trained using datasets with labeled examples. The model learns how to predict the label from the features. However, not every feature in a dataset has predictive power. In some instances, only a few features act as predictors of the label. In the dataset below, use price as the label and the remaining columns as the features.

Features										Label
color	miles	height	gearbox	make_model	length	tire_size	wheel_base	weight	year	price

Which three features do you think are likely the greatest predictors for a car's price?

Color, height, make_model.

Make_model, year, miles.

A car's make/model, year, and miles are likely to be among the strongest predictors for its price.

Correct answer.

Miles, gearbox, make_model.

The gearbox isn't a main predictor of price.

Try again.

Tire_size, wheel_base, year.

Tire size and wheel base aren't strong predictors for a car's price.

Try again.

Supervised and unsupervised learning

Based on the problem, you'll use either a supervised or unsupervised approach. For example, if you know beforehand the value or category you want to predict, you'd use supervised learning. However, if you wanted to learn if your dataset contains any segmentations or groupings of related examples, you'd use unsupervised learning.

Suppose you had a dataset of users for an online shopping website, and it contained the following columns:

last_purchase_amt	years_since_signup	num_purchases	num_coupon_usage	education	marital_status	income
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If you wanted to understand the types of users that visit the site, would you use supervised or unsupervised learning?

Supervised learning because I'm trying to predict which class a user belongs to.

Unsupervised learning.

Because we want the model to cluster groups of related customers, we'd use unsupervised learning. After the model clustered the users, we'd create our own names for each cluster, for example, "discount seekers," "deal hunters," "surfers," "loyal," and "wanderers."

Correct answer.

Suppose you had an energy usage dataset for homes with the following columns:

sq_ft	loc_lat_long	year_built	kwhrs_peryear
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What type of ML would you use to predict the kilowatt hours used per year for a newly constructed house?

Unsupervised learning.

Supervised learning.

Supervised learning trains on labeled examples. In this dataset "kilowatt hours used per year" would be the label because this is the value you want the model to predict. The features would be "square footage," "location," and "year built."

Correct answer.

Suppose you had a flight dataset with the following columns:

departure_airport	destination_airport	date	time_of_day	airline	fuel_price	airplane_ticket_cost
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If you wanted to predict the cost of an airplane ticket, would you use regression or classification?

Classification

A classification model's output is a discrete value, normally a word. In this case, the cost of an airplane ticket is a numeric value.

Try again.

Regression

A regression model's output is a numeric value.

Correct answer.

Based on the dataset, could you train a classification model to classify the cost of an airplane ticket as "high," "average," or "low"?

No. It's not possible to create a classification model. The `airplane_ticket_cost` values are numeric not categorical.

No. Classification models only predict two categories, like `spam` or `not_spam`. This model would need to predict three categories.

Yes, but we'd first need to convert the numeric values in the `airplane_ticket_cost` column to categorical values.

It's possible to create a classification model from the dataset. You would do something like the following:

Find the average cost of a ticket from the departure airport to the destination airport.

Determine the thresholds that would constitute "high," "average," and "low".

Compare the predicted cost to the thresholds and output the category the value falls within.

Correct answer.

Training and evaluating

After we've trained a model, we evaluate it by using a dataset with labeled examples and compare the model's predicted value to the label's actual value.

Select the two best answers for the question.

If the model's predictions are far off, what might you do to make them better?

Try a different training approach. For example, if you used a supervised approach, try an unsupervised approach.

Retrain the model, but use only the features you believe have the strongest predictive power for the label.

Retraining the model with fewer features, but that have more predictive power, can produce a model that makes better predictions.

2 of 2 correct answers.

Retrain the model using a larger and more diverse dataset.

Models trained on datasets with more examples and a wider range of values can produce better predictions because the model has a better generalized solution for the relationship between the features and the label.

1 of 2 correct answers.

You can't fix a model whose predictions are far off.

It's possible to fix a model whose predictions are off. Most models require multiple rounds of training until they make useful predictions.

Try again.

You're now ready to take the next step in your ML journey:

[Machine Learning Crash Course](#). If you're ready for an in-depth, hands-on approach to learning more about ML.

[Problem Framing](#). If you're looking for a field-tested approach for creating ML models and avoiding common pitfalls.

[People + AI Guidebook](#). If you're looking for practical guidance for designing human-centered AI products.

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